FAST EventFlow D4 Prototype Report

CS3014 Applied Human Computer Interaction | D08 - Shmeezki & Mahid

Prepared for instructor review | 4 November 2026 | Build 1.0

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This package documents a working status-first approval prototype, its grayscale structure, design system, scope and verification. It connects the D2 requirements and D3 concept to the actual implementation and provides forms for team-owned activity records.

The build passes the recorded automated workflow checks. The IA is team-proposed; feature-sort records, team accessibility walkthroughs, critique notes, reflections, captioned team narration and the instructor checkpoint are completed through the accompanying forms and recording instructions.

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Research and implementation traceability

Source basis

D1A establishes assumption A1 and the research question about how leads find out a request's status and act during silence. D1B records seven interviews and discloses a student-weighted sample: six society leads, one mentor and no One Stop or student-affairs officials. D2 defines 35 evidence units, the affinity synthesis, Hira, the point of view and five measurable requirements. The supplied wall photographs corroborate that earlier synthesis; they are not a D4 feature sort or a new study. No participant identities, real society names or raw research photographs are republished in the prototype.

D3 selects the status-first portal and explains the additions of SF1 submission checking, AM5 reasoned decisions and MS5 overdue escalation. This is the mapping used in the build. The current project specification treats the selected concept as the implementation baseline; D3's earlier “team to confirm” wording is not evidence of a later meeting.

The authoritative D4 brief requires a card-sort-derived IA and the team's own accessibility observations. This package supplies a proposed IA, activity pack and recording forms; prepared forms are not substituted for activity records. It also requires original team narration, captions and a live instructor checkpoint, which cannot be inferred from an automated build.

Requirement matrix

Requirement and D2 criterion	D2 basis and prototype screens	Implemented mechanism	Verification and status
R1: at least 90 percent identify stage within 30 seconds unassisted	P1-01, P1-02, P3-02, P4-01, P5-01; W1, W3, W4, W8	Current stage, holder, timestamp and history in the same request context	Browser detail and history checks pass; timed comprehension criterion remains an evaluation target
R2: at least 80 percent of first-timers submit a complete request on first attempt without help	P1-04, P5-03; W2, W6, W7, W8, W9, W10	Required instructions, custom validation, retained drafts, correction and confirmation	New and corrected submissions pass browser checks; first-time success rate is not established by code tests
R3: at least 90 percent explain refusal in their own words	P5-03, P6-05, S1-02; W5 and W6	Outcome and meaningful reason together, correction when permitted	Refusal and changes-needed reasons verified visible; comprehension criterion remains an evaluation target
R4: at least 85 percent classify one normal and one overdue request	P2-03, P4-03, P6-03; W1, W3, W4	Text labels, expected date, elapsed days and simulated escalation disclosure	Fixed-date scenarios verified; human classification rate remains an evaluation target
R5: keyboard status task, screen-reader comprehension, zero contrast failures and non-colour status	Course requirement and context of use; all screens	Native controls, skip link, focus management, semantic structure, error associations and text status	Automated keyboard assertions and axe scans pass within scope; team walkthrough and spoken-output records use the supplied form

Evidence register

Supplied source	Verified reference used
D1A research plan	Assumption A1 and neutral behavioral research question; sampling and anonymity commitments
D1B fieldwork	Seven-session evidence base and reflection on staff-side access and limitations
D2 Synthesis and Requirements	Pages 2-4 cluster register and needs; page 5 persona; pages 6-9 task framing and measurable requirements; pages 10-17 process evidence
D3	Pages 8-10 selected concept, storyboard and rationale; page 11 open risks

Supplied source	Verified reference used
Official D4 brief	Pages 1-2 Part A and B; page 3 video and AI rules; page 4 reflection and continuity; pages 4-5 checklist
Original D4A HTML	Eight screen compositions and their hierarchy and Gestalt annotations
CLAUDE.md	Team metadata, current specification, house style and open-work boundaries

The report uses team names and IDs for authorship only. Event requests, societies and venues in the executable demonstration are fictional, independent of the research participants.

Information architecture and card sorting

Proposed structure and rationale

The proposed navigation is hub-and-spoke. My Requests is the recurring hub: a society lead can identify a request, open its stage and return to the list. New Request is a distinct entry for first-time submission; Help explains stages and correction without introducing an administrative menu. Request history and decision reasons belong within the relevant request because their meaning depends on its ID.

This is the team's proposed structure, grounded in the D2 affinity synthesis. D2 affinity grouping concerns research observations; the D4 feature card sort concerns how people organize portal functions. These are separate artifacts. The supplied results form records the D4 activity when completed, and the navigation can then be revised against that evidence.

D2 C1 and C2 support making stage, holder and update visible together. C3 supports ownership and history within a request. C4 supports field guidance, reasoned decisions and correction. C5 supports expected dates; C6 corroborates planning risk but does not justify a separate logistics module. S1-06 is a suggested solution and is not used as a need. The proposed structure keeps the prototype focused on visibility and action rather than replacing the entire approval process.

Content inventory and feature cards

The numbered labels below are the printable card pack. Give participants the labels and descriptions without the proposed navigation column. Proposed categories express the current design hypothesis, not recorded sorting results.

Card	Label and content	Proposed location	Requirement
C01	Find my submitted event requests	My Requests	R1
C02	See the current approval stage	Request detail	R1
C03	See which office holds the request	Request detail	R1
C04	See when the request was last updated	Request detail	R1
C05	See the expected review date	Request detail	R4
C06	Identify an overdue request	My Requests and detail	R4

Card	Label and content	Proposed location	Requirement
C07	Read a decision and its reason	Request detail	R3
C08	Read the approval history	Request detail	R1 and R3
C09	Correct a request needing changes	Request correction	R2 and R3
C10	Save unfinished form entries	New or correction form	R2
C11	Check which fields need attention	Form error state	R2 and R5
C12	Submit a new event request	New Request	R2
C13	Confirm that submission was received	Confirmation	R2
C14	Learn what approval stages mean	Help	R2
C15	Find keyboard and form guidance	Help	R5
C16	Understand an overdue escalation label	Overdue detail and Help	R4

Navigation diagram

See Information-architecture.svg for the grayscale diagram. The top navigation contains My Requests, New Request and Help. My Requests branches to on-time, overdue and decision details. Changes needed branches to Edit and resubmit. New and correction forms share validation and confirmation patterns. Each detail or confirmation offers a return to My Requests.

The primary path is W1 > W4 > W5 > W6 > W8 > W1. W4 > W5 represents a staff update, explicitly simulated by a demo control. The error path is W6 > W7 > correction > W8; creation follows W1 or W2 > W9 > W7 if invalid > W9 correction > W8. W10 Help is available globally. Browser Back and direct hash routes work alongside visible navigation links.

Card sorting session pack

Use the separate printable HTML pack. Allow approximately twenty minutes: two minutes for the task and consent, eight for independent grouping, five for naming and explaining groups, and five for discussion. Invite appropriate society leads or clearly identified stand-ins. Use anonymous IDs and role categories; avoid identifying society or staff names. Obtain agreement before recording any notes or images.

Read this neutral instruction: “These cards describe things an event-request portal might let you do. Group things where you would expect to find them together. Make as many or as few groups as you need. Name each group. Put confusing cards aside or rename them. There are no correct answers; explain the choices that matter to you.”

Shuffle all sixteen cards. Start with an open sort; do not reveal My Requests, New Request and Help in advance. Record each person's grouping before discussion. Ask where they would look to find an overdue request, understand a refusal, correct it and begin a new submission. If comparing the proposed navigation afterward, label that separately as a closed-sort check. Do not treat agreement in discussion as independent participant agreement.

Analysis and decision rules

Keep the original placements, proposed group names, ambiguous cards and explanations. Count actual placements before calculating any agreement; report the denominator and method. For a very small sort, prefer qualitative disagreement and reasoning over percentages. Decide whether recurring group names improve navigation; inspect any split between decision history and request details. Log the final navigation decision, affected screens, rationale and subsequent prototype changes.

The accompanying results template has no prefilled participant outcomes. Use it for the activity rather than relabelling the D2 affinity wall. Add the actual date, IDs, group labels, card membership and decisions, then replace the proposed IA description with the supported result.

Requirements to navigation

Requirement	Navigation mechanism	Why it belongs there
R1	Dashboard > request detail > history	Locate the same request and inspect its current owner without another person
R2	New Request or Changes needed > form > errors or confirmation	Preparation and recovery remain beside the task
R3	Dashboard > decision reason > correction if allowed	The outcome and next action share the request context
R4	Dashboard status > expected date and overdue detail	Compare normal and late requests before inspecting timing evidence
R5	Global skip link and Help; native controls across routes	Preserve a consistent keyboard entry and meaningful reading order

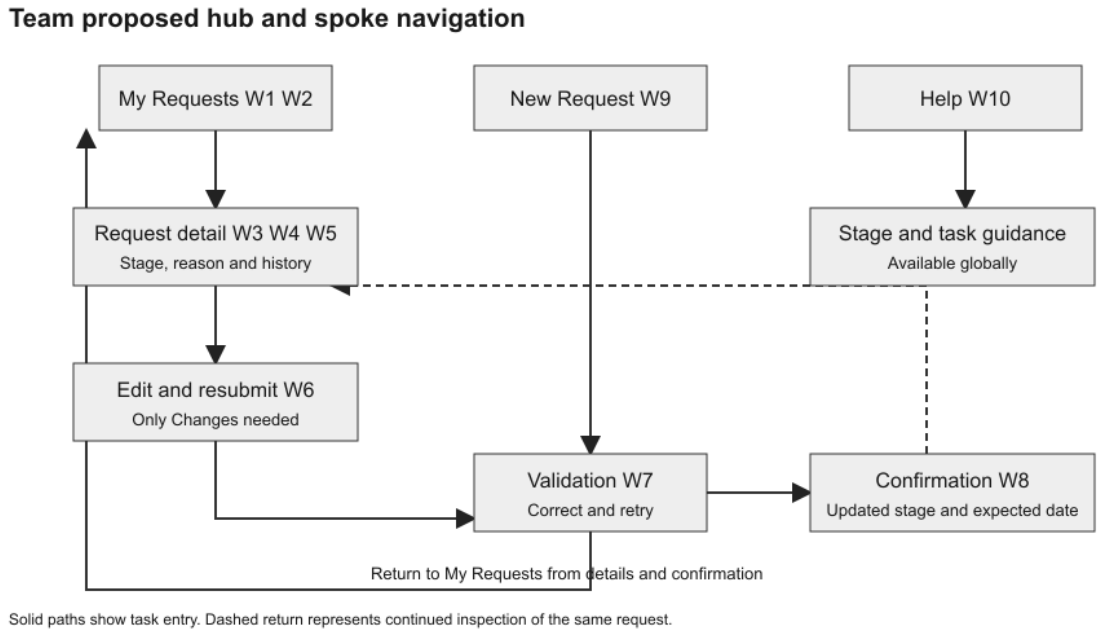


Figure 1. Proposed navigation; separate results form records subsequent feature-sort decisions.

Wireframe inventory and review

The original supplied eight-screen HTML is preserved byte for byte in sources. The Part A working copy keeps its grayscale frames, status treatment, navigation map and annotations. It adds navigation actions and the

missing creation and Help destinations. Fields in Part A are explicitly static sketches; Part B supplies real form behavior.

Screen mapping and hierarchy

ID	Screen and behavior	Hierarchy and Gestalt principle	Requirement
W1	My Requests with links to request details	Title then status then metadata; proximity groups each request	R1 and R4
W2	Empty dashboard leading to W9	Empty headline then explanation and primary action; proximity groups the next step	R2
W3	On-time detail with office, timestamp and history	Title and stage precede timing metadata; continuity links the stage sequence	R1 and R4
W4	Overdue detail and simulated escalation	Exception precedes metadata; common region isolates the overdue explanation	R4
W5	Decision and stated reason, with correction action where allowed	Reason precedes action; similarity reuses the exception treatment	R3
W6	Edit and resubmit with reviewer reason above fields	Label, field and message stay together through proximity	R2
W7	Shared validation error state and recovery	Summary precedes affected fields; similarity makes error marks recognizable	R2 and R5
W8	Submission or resubmission confirmation	Result then updated stage and expected date; similarity repeats the request structure	R1 and R2
W9	Added new request form	Preparation then required fields then action; proximity groups each field	R2
W10	Added Help	Task-based headings; similarity preserves stage and action vocabulary	R2 and R5

Substantive modifications

1. Added W9 and W10 so New Request and Help have defined destinations in both parts. Added a text flow beside the preserved diagram for creation, shared errors and return routes.

2. Connected request titles and action buttons to relevant wireframe tabs. Retained arrow-key, Home and End tab selection. Added return actions on the original frames and replaced dead history buttons with expandable sketches.

3. Added hierarchy and Gestalt notes for W2, W9 and W10. Preserved existing W1 and W3-W8 annotations. These are design rationales for team review, not participant feedback.

4. Labelled escalation and availability as illustrative. Removed the working copy's unsupported promise that specific real venues were free. The original remains unchanged for comparison.

5. Changed field-sketch labels to text descriptions rather than implying association with actual inputs. Added a main landmark and a focusable skip destination. Darkened muted footer text from #8A8A93 to #62626B.

6. Distinguished the original October snapshots from the functional build's fixed 4 November clock. The event conflict in Part B is on 22 November, allowing future-date correction in a current demonstration.

Review conclusions

The original structure covered status, delay, decision and correction, but New Request and Help were destinations without screens, several frame buttons were inert and the fields were sketches. The extension preserves that structure while making the path inspectable. The Part B form intentionally allows submission attempts with missing fields so W7 can explain errors; it disables submission during processing to prevent duplicate sends. This resolves the project specification's tension between disabling empty forms and demonstrating missing-field errors.

The reason block stays visible while editing. Correction preserves the request ID, society, plan and history. Decision outcomes are terminal for overdue calculation; a refused request is not mislabelled overdue. W8 offers both updated details and the hub, avoiding a confirmation dead end. Usability criteria remain evaluation targets, not consequences of this review.

Accessibility inspection and walkthrough forms

Evidence boundaries

Use this form for the team's Part A walkthrough and Part B re-check on the actual delivered version. Automated evidence is recorded separately in Part B Testing and evidence/test-results.json. The official brief requires the team's own keyboard and screen-reader findings; generated inspection suggestions and automated assertions do not replace those logs.

Record artifact path and SHA256, date, tester, browser/version, operating system, assistive technology/version, viewport and input method. Record a finding only after observing it. Mark each issue fixed, deferred or pending verification, with a retest reference. Severity: critical blocks the core task; major obstructs completion; moderate adds avoidable effort; minor is a limited presentation issue.

Perceivable

Inspect W1-W5 and W8 at default size, 200 percent zoom, a narrow viewport and grayscale. Confirm that every chip has a word, completed stages have a mark and current stages have text. Compare normal and overdue requests without relying on colour. Inspect form help and errors for readable text and spacing. Check text contrast against its actual background, including hover, focus, selected navigation, alerts and disabled controls. For WCAG 2.1 AA use 1.4.1 Use of Color, 1.4.3 Contrast Minimum, 1.4.10 Reflow and 1.4.11 Non-text Contrast as applicable. Normal text requires at least 4.5:1; large text and relevant non-text indicators require 3:1.

Expected: status remains unambiguous in grayscale; text and required boundaries are distinguishable; content remains readable without clipping. Axe's contrast result covers inspected rendered states only; verify any incomplete result and additional zoom/interaction states manually.

Operable

On Part A, begin at the top using only the keyboard. Activate the skip link, enter the screen index, use arrow keys, Home and End, and Tab into the chosen panel. Follow request, correction, error and return actions. Record any unexpected jump, unreachable control or trapped focus. Repeat on Part B from W1 through W3, W4, W5, W6, W7 and W8, then through W2 and W9. Use Tab and Shift+Tab in both directions, Enter on links, Space and Enter on buttons, and native select and date controls. Check that focus is clearly visible and is not hidden by content.

Expected: no mouse is needed; only the selected wireframe tab enters the normal tab order; main-route changes focus the heading, errors focus the summary and summary links focus the relevant input. The skip link focuses main content without changing the application route. Check 2.1.1 Keyboard, 2.1.2 No Keyboard Trap, 2.4.1 Bypass Blocks, 2.4.3 Focus Order, 2.4.6 Headings and Labels and 2.4.7 Focus Visible.

Understandable

Open W9, submit it empty, and record the message and focus destination. Fill the title and note, leave date and venue empty and attempt resubmission in W6. Confirm that nothing is sent, entered values remain, each error identifies its field and offers a correction, and a second valid attempt reaches confirmation. Attempt the original conflicted date/venue pair to check that the reason remains actionable. Cancel and reopen the form to inspect draft behavior. Check required instructions before submission, not only after an error.

Expected: labels are persistent; mandatory fields are identified; errors do not depend on a red border; W7 states what did not happen; W8 identifies the request, current office and expected date. Inspect 3.2.3 Consistent Navigation, 3.2.4 Consistent Identification, 3.3.1 Error Identification, 3.3.2 Labels or Instructions and 3.3.3 Error Suggestion.

Robust

Use Narrator with Edge or NVDA with a supported browser and log the actual words heard. Navigate by landmarks and headings, then inspect a card's link name, stage list, decision reason and form labels. Trigger validation; check announcement and reading order from summary to field. Resolve errors and submit; check whether loading and confirmation are conveyed without unnecessary repetition. Inspect aria-invalid and aria-describedby associations, the polite status region, current-page and current-step attributes and expandable history. Check 1.3.1 Info and Relationships, 4.1.2 Name Role Value and 4.1.3 Status Messages.

Expected: each input has a meaningful name and associated help/error; status updates are available to assistive technology; redundant or meaningless announcements are recorded for correction. A programmatic DOM check cannot establish the actual spoken experience.

Issue record template

Issue ID and screen ID: _____

POUR principle and WCAG criterion: _____

Artifact version, tester, date and environment: _____

Reproduction steps and input method: _____

Actual observed behavior or exact spoken announcement: _____

Expected behavior and impact: _____

Severity and recommended correction: _____

Status fixed / deferred / pending verification: _____

Fix reference, retest steps, result and remaining concern: _____

Engineering corrections recorded separately

Screen	Principle	Evidence and correction	Status
Part B all	Operable	The browser test initially found the skip link changed the hash to main, causing an unavailable-route screen. The link now focuses main without routing. Repeated keyboard assertion passes.	Fixed in code and programmatically verified
Part A W1-W10	Operable	Code review found inert frame actions. Actions now select destination tabs; a browser assertion follows W1 to W9.	Fixed; team walkthrough pending verification
Part A all	Perceivable	Muted footer text was darkened in the working copy; automated scans run on every panel.	Fixed; additional visual conditions pending verification
Part B W6-W7	Understandable and Robust	Error summary focus, preserved note, invalid-field associations and summary-link focus are asserted in the browser suite.	Implemented and programmatically verified

Deferred and pending re checks

No observed team walkthrough issue is entered in this prepared form. Team keyboard and spoken-output records, zoom/reflow checks and assessment of automated incomplete results remain pending verification. Backend permissions, real university deadlines and real notification behavior are outside the prototype and must not be interpreted as accessibility fixes. Revisit any team-deferred Part A issue on Part B before the instructor checkpoint.

Reference: W3C WCAG 2.1 https://www.w3.org/TR/WCAG21/ and WAI explanations of Error Identification and Status Messages. These references guide inspection; this document makes no overall conformance claim.

Card sort record

Activity status: team to complete. Artifact reviewed: proposed D4 feature navigation, version 1.0. This form records a feature card sort, not the D2 affinity synthesis.

Session date and duration: _____

Facilitator and note taker: _____

Participant or stand-in IDs and role categories: _____

Consent and recording arrangements: _____

Method and deviations from the pack: _____

Original placements

Copy one row per independently created group. Keep original wording; keep individual results separate before discussion. Use Card-sort-placements.csv for machine-readable records.

Anonymous ID	Group name as given	Card IDs in this group	Explanation or ambiguity
To complete	To complete	To complete	To complete

Interpretation and revision

Cards repeatedly grouped together and underlying counts: _____

Disagreements or cards set aside: _____

Preferred labels and supporting explanations: _____

Decision about My Requests, New Request and Help: _____

Changes to the IA and affected wireframe IDs: _____

Remaining ambiguity and follow-up task: _____

Evidence filename or consented image reference: _____

Team reviewer, review date and acceptance: _____

Studio critique record and checklist

Session date, duration and location: _____

Participants or reviewer role categories: _____

Prototype version and screen/task reviewed: _____

Feedback as recorded, with the speaker's intended meaning: _____

Issue identified and supporting observation: _____

Priority and reason: _____

Team decision and owner: _____

Changes made, affected screens and version: _____

Retest or review evidence and remaining concerns: _____

This is a prepared recording form, not minutes of a completed critique. Use one issue record per distinct concern and separate a reviewer's observation from the team's interpretation.

Practical review sequence

1. Ask the reviewer to identify the stage and responsible office of #E-031, then compare it with #E-024. Record confusion over On time, Overdue, expected review date or last update.

2. Show the Refused example. Ask the reviewer to explain its reason and likely next step in their own words; do not prompt the explanation.

3. Show Changes needed and correction. Inspect whether the reason remains visible and whether the editable fields match what is requested.

4. Trigger W7 with date and venue empty. Inspect summary placement, retained input, field messages and the recovery path to W8.

5. Try the empty dashboard and New Request. Check whether the action label and required information are understandable without a tour.

6. Inspect keyboard focus, non-colour statuses, narrow layouts and Help. Record accessibility observations in the separate audit log.

7. Agree a small prioritized change list, assign owners and retain deferred concerns. Do not convert critique comments into measured usability success rates.

Functional prototype and design system

Core task and scope

FAST EventFlow implements one complete task: find out where a submitted event request stands and act on what you find. A society lead opens My Requests, inspects stage, office, last update and deadline, reads a reasoned decision, corrects a Changes needed request, recovers from field errors and sees the updated state after resubmission. New Request also works from the hub and empty state. Help explains the same vocabulary used in the interface.

The repository initially contained only a README and one initial commit. The implementation therefore follows the supplied project preference: one self-contained HTML file with inline CSS and vanilla JavaScript. There is no build step, framework, backend or runtime dependency. Opening index.html through file:// is the tested launch path. Development-only Playwright and axe dependencies support repeatable verification.

The vertical slice deliberately excludes authentication, university routing, actual staff review, document uploads, venue booking, event execution and resource procurement. An event-plan text field stands in for supporting documents. This keeps correction, status and recovery deep enough to inspect while making scope boundaries visible. Actual university submission rules and policy validity are not established by this prototype.

State and dates

Each request contains ID, title, society, event date, preferred venue, event plan, optional note, original submission timestamp, current stage, holder, last update, expected-by date, decision, reason, decision timestamp, correction deadline and approval history. Drafts are stored separately by request ID or new. Edits preserve the original ID, submission time, society, plan and history; valid resubmission clears the old decision, appends a history entry and returns to Department review.

The clock is fixed at 4 November 2026 and is displayed on every screen. Overdue is derived for undecided requests only, after the expected calendar date has passed. #E-024 is two days past 2 November; #E-031 is on time until 6 November. Outcomes are labelled as outcomes even if their old expected date has passed. Resubmission sets an illustrative review date of 7 November; a new request goes to the Faculty Mentor with expected review by 8 November. These are demonstration assumptions, not university service commitments.

Local storage uses the versioned key fast-eventflow-d4-v1. Invalid saved data falls back to samples with a notice; unavailable storage retains changes in the current tab and warns the user. No account or cross-device synchronization exists. Demo controls explicitly replace local sample requests and drafts. Staff simulations append a history entry and are visibly identified as simulated.

State transitions and feedback

Routes use hashes for dashboard, request details, correction, creation, confirmation and Help. Missing or no-longer-editable requests show an explanation and a hub link. Route changes focus the main heading and update the document title. A polite status region reports saved drafts, filters, errors, loading and successful actions.

New submission requires title, society, future event date, preferred demo venue and event plan. Correction requires title, date and venue; it carries forward society and plan. The optional note is preserved. A missing field, past date, unrecognized venue or unresolved original date/venue pair prevents submission. Field values remain in place, an error summary receives focus, error links lead to fields, and aria-describedby connects help and error messages. The summary says that nothing was sent.

Submit remains available on an incomplete form so the required error scenario can be demonstrated. During a valid send, buttons are disabled and duplicate sends are blocked. The loading label appears after 200 milliseconds during a deliberately simulated 650 millisecond local send delay. On completion, the confirmation states the same request ID, new stage, office and expected date. Input events retain drafts locally; Save draft gives explicit feedback; Cancel leaves the draft available.

Built and simulated

Capability	Implementation boundary
Navigation, filtering, empty and unavailable states	Real client-side behavior
Required-field validation, preservation and correction	Real client-side behavior
Request creation, resubmission and history updates	Real in-browser state transitions
Overdue derivation and outcome labels	Real calculation against the visible fixed demo clock
Focus, labels, error associations and live region	Implemented browser behavior; spoken experience requires team review
Staff decisions and stage updates	Simulated through instructor controls and fictional samples
Escalation	Illustrative status text; no alert or message is sent
Submission waiting time	Deliberate local delay to expose loading and disabled states
Authentication and authorization	Absent; no real user identity or private university account
Venue choices and expected dates	Fictional values; no availability or deadline service
Supporting documents and bookings	Outside the slice; no upload or booking control is presented

Palette and typography

Token	Value	Use
Neutral background	#F5F6F8	Page
Neutral surface	#FFFFFF	Cards, forms and controls
Neutral text	#18202D	Main text and focus outlines
Neutral secondary	#505B6B	Metadata and help
Neutral line	#B7BFCA	Structural separators
Primary accent	#4338A5	Primary action and current navigation
Success	#176044 on #EDF7F1	Confirmation and completed stage
Warning	#794400 on #FFF4DF	Overdue and changes needed
Error	#A32332 on #FFF0F1	Refusal and field errors
Information	#245B83 on #EFF6FC	Neutral status and notices

Use Segoe UI with Arial and sans-serif fallbacks; weights 400 and 600. Screen headings are 32px, reducing to 28px on narrow screens; section and row headings are 20px; body is 16px with metadata at 14px. The three hierarchy tiers are screen title, section/row title and body/metadata. Line height is 1.55 for body text. Layout spacing uses 4, 8, 12, 16, 24 and 32px, with larger empty-state padding for emphasis.

Text carries every status independently of semantic colour. Progress has labelled stages, completed-stage check marks and a current-step label. Overdue badges use an additional border weight and elapsed days. Fields have persistent labels, a visible required marker and separate error text. Buttons have at least 44px height and a visible 3px focus outline with offset.

Components and states

Component or state	Concrete use
Request card	ID, fictional society, linked title, text badge, stage list and timing metadata
Status badge	On time, Overdue with elapsed days, Approved, Changes needed or Refused
Progress indicator	Ordered list with aria-current step and text labels
Form field	Label, help, native input or select, associated error
Primary and neutral buttons	Main submit action and draft or demo actions
Callout and confirmation	Reason, overdue explanation, validation summary and successful receipt
Default	Dashboard cards and enabled controls
Hover and focus	Background feedback and visible offset outline
Active	Button movement on press
Disabled	Submission buttons while a valid send is processing
Loading	Initial request loading and delayed Submitting label
Empty	No requests yet, plus a working creation action
Error	W7 summary, field messages and retained values
Success	W8 receipt and polite saved-state notice

Remaining design risks

A portal can show an inaccurate stage if staff do not maintain it. This build demonstrates staff updates through clearly labelled controls and retains timestamps, but it cannot establish real update ownership. Stage deadlines must be agreed with actual offices. Refusal comprehension, first-time form completion and classification of normal versus overdue remain measurable evaluation questions. Shared-computer privacy and real role permissions require a backend design beyond this local slice.

Automated verification record

Actual run

Run date: 4 November 2026. Browser: Microsoft Edge 156.0.4314.58, headless through Playwright 1.62.1. Axe-core: 4.10.3. Mode: local file:// execution. The prototype required no server, build, network or

university connection. The record timestamp is 2026-11-04T13:58:25.350Z (UTC); client timezone is Asia/Karachi.

Result: 18/18 browser scenarios passed. All 24 axe snapshots reported zero violations under wcag2a, wcag2aa, wcag21a and wcag21aa tags. 13 snapshots contain incomplete contrast checks, detailed below. A zero-violation scan is not a full WCAG conformance finding.

Prototype SHA256: 2feb7d075ce67ab5a7948a8442299cdaa564704cda927cb00781880f8ef8e221

Commands actually used

The workspace dependency loader supplied the bundled runtime. In PowerShell, set runtime to the returned dependencies directory, then run these commands from the repository:

$env:NODE_PATH = "$runtime\node\node_modules"

$env:AXE_PATH = (Resolve-Path 'work\axe.min.js').Path

& "$runtime\python\python.exe" -X utf8 tools\extend-wireframes.py

& "$runtime\node\bin\node.exe" tools\check.cjs

& "$runtime\node\bin\node.exe" tests\verify.cjs

The initial setup downloaded axe-core 4.10.3 from https://cdnjs.cloudflare.com/ajax/libs/axe-core/4.10.3/axe.min.js to work/axe.min.js. No prototype runtime dependency was installed. On a normal developer installation, npm install supplies the pinned development packages; npm run check and npm test invoke the same scripts. There is no separate build or lint configuration to run.

Syntax and duplicate static-ID checks passed for both HTML artifacts. The first pass exposed a skip-link routing defect; the implementation was corrected and the keyboard assertion passed on the recorded repeat. A filter test initially requested a control in the wrong empty-data fixture; the fixture was corrected. The static ID checker initially included script template strings and data-id attributes; it now checks actual static markup. The final evidence reflects the corrected checks and application.

Browser scenarios

Scenario	Result
W1 dashboard loads four requests and initial loading ends	PASS
W3 on-time details include stage, holder, update and history	PASS
W4 overdue derives two days from fixed clock and labels simulation	PASS
W5 refusal shows meaningful reason and no edit action	PASS
Simulated staff changes produce reason and correction action	PASS
W6 form retains original fields, drafts persist across reload	PASS
W7 invalid submission blocked; field messages and focus; input retained	PASS
Error summary link focuses field without changing route	PASS

Scenario	Result
W8 valid resubmission shows loading, disabled state and updated stage	PASS
Empty state and New Request complete end to end	PASS
Help, missing request and all outcome states remain navigable	PASS
Filters and no-results recovery work	PASS
Programmatic keyboard path: skip link, navigation, focus and reverse Tab	PASS
Responsive view and grayscale retain text status; no horizontal overflow	PASS
Markup escapes user text	PASS
Corrupt storage recovers; blocked storage degrades visibly	PASS
Part A ten panels and keyboard screen selection work	PASS
No browser JavaScript exceptions	PASS

Automated accessibility snapshots

Screen or state	Reported violations	Incomplete checks
W1 dashboard	0	None
W3 on time	0	None
W4 overdue	0	None
W5 refused	0	None
W5 changes needed	0	None
W6 edit	0	color-contrast (1 nodes)
W7 validation	0	color-contrast (1 nodes)
W8 resubmitted	0	None
W2 empty	0	None
W9 new	0	color-contrast (2 nodes)
W8 new submission confirmation	0	None
W10 help	0	None
W5 approved	0	None
W1 mobile grayscale	0	None

Screen or state	Reported violations	Incomplete checks
Part A W1	0	color-contrast (25 nodes)
Part A W2	0	color-contrast (25 nodes)
Part A W3	0	color-contrast (25 nodes)
Part A W4	0	color-contrast (25 nodes)
Part A W5	0	color-contrast (25 nodes)
Part A W6	0	color-contrast (25 nodes)
Part A W7	0	color-contrast (25 nodes)
Part A W8	0	color-contrast (26 nodes)
Part A W9	0	color-contrast (24 nodes)
Part A W10	0	color-contrast (24 nodes)

Incomplete contrast checks arise in form/date controls and SVG wireframe text. They are retained in the raw JSON for human assessment rather than reported as passes. The following calculated palette ratios supplement the scans; they do not prove every composited or interactive state.

Foreground and background pair	Ratio	Threshold used
Main text (#18202D / #FFFFFF)	16.36:1	4.5:1 normal text
Secondary text (#505B6B / #FFFFFF)	6.89:1	4.5:1 normal text
Accent on white (#4338A5 / #FFFFFF)	8.95:1	4.5:1 normal text
Success (#176044 / #EDF7F1)	6.87:1	4.5:1 normal text
Warning (#794400 / #FFF4DF)	7.30:1	4.5:1 normal text
Error (#A32332 / #FFF0F1)	6.68:1	4.5:1 normal text
Information (#245B83 / #EFF6FC)	6.64:1	4.5:1 normal text
Part A muted labels (#62626B / #FFFFFF)	6.04:1	4.5:1 normal text
Focus on page (#18202D / #F5F6F8)	15.13:1	3:1 non-text
Input boundary (#707B8D / #FFFFFF)	4.28:1	3:1 non-text

Scope and re check

The suite exercises visible states, real DOM events, keyboard keypresses, focus destinations, local state persistence, error recovery, a narrow grayscale view and escaped user text. It checks that the browser reports no JavaScript exceptions. It does not measure usability, test real administrative decisions or establish actual speech output. Additional browser/assistive-technology combinations, 200 percent zoom, forced colours and team-owned walkthrough evidence use the Part A audit procedure.

Evidence files: test-results.json and seven PNG screenshots in evidence/. Keep the raw record with the submitted build and repeat verification after changing executable HTML. Generated documentation and tests do not establish the D2 success percentages.

AI disclosure and team reflection

AI use note for team review

OpenAI Codex assisted with reading the supplied project materials, repository discovery and setup, extending the existing grayscale wireframes, implementing the vanilla JavaScript prototype, drafting documentation and forms, generating the consolidated report and preparing the video script. It also authored and ran the browser integration suite, axe scans, syntax checks and document layout checks recorded in the evidence files. Playwright and axe are verification tools, not participant simulators.

The supplied original wireframes and previous deliverables predate this work; their own disclosures remain their provenance records. Codex used the existing D2 evidence IDs and synthesis as context and did not create a replacement affinity grouping. The D4 navigation is proposed from that context. The original wireframe artifact is preserved for comparison with the extended working copy.

Review status: AI implementation and programmatic verification are recorded. The team should replace this sentence with the specific code, wording and design decisions they reviewed or modified, together with reviewer and date. No team review, attendance, critique feedback, participant outcome, spoken-output finding or personal reflection is attributed without an actual record.

The official brief allows AI assistance for wireframes and the build with disclosure. It reserves feature-sort evidence for actual users or stand-ins and accessibility findings for the team's own walkthroughs. The prepared forms support those steps; generated text is not evidence that they occurred. The recording script is a rehearsal aid. Original narration and the captioned MP4 are team deliverables.

Team review record

Reviewer name and date: _____

Code and design sections personally examined: _____

Changes made and reasons: _____

Tests personally repeated and observed results: _____

Parts the reviewer can explain at the checkpoint: _____

Official reflection prompts

Answer each prompt in one or two sentences in your own words. These are the two prompts from the brief; the response spaces intentionally contain no generated personal reflection.

1. What did you keep as your own thinking versus what came from AI?

Team response: _____

_____

2. How did it feel to flag your AI use here?

Team response: _____

_____

Five minute recording script and shot list

Recording setup

Required output filename: D08-Shmeezki&Mahid-D4-Prototype.mp4. Target duration 4:55 to leave export headroom below five minutes. Record the actual index.html build 1.0 with original team narration and captions. Use 1080p capture, a readable browser zoom and an uncluttered desktop. Keep the fixed-date banner visible. Rehearse once so interactions match the spoken explanation. Captions should transcribe the final spoken narration, with readable line breaks and timing; this script is not a substitute for a checked caption track.

Before recording, expand Instructor demo controls and Load sample requests. Close the controls except when showing a staff simulation. Keep the alternate path on the same submitted build. After exporting, play the MP4 locally, check duration, speech/caption synchronization and interface legibility, then check sharing permissions and keep a local backup. All team members must be ready to explain the live build.

0 00 to 0 30 User task and problem

Shot: W1 My Requests. Point at the request cards and their office/update lines; do not begin with code or implementation details.

Narration: “Our user is Hira, a society president responsible for getting an event approved. In our earlier research, leads described visiting offices or contacting staff to find out where a request stood. Silence did not tell them whether it was moving or stalled. FAST EventFlow focuses on one task: find the current stage of a submitted request, understand the wait, and act on a recorded decision.”

0 30 to 1 00 Requirements and design response

Shot: Point to stage, holder, expected date and badge on the dashboard. Briefly open Help if needed, then return. Keep the interface readable.

Narration: “Our five requirements cover stage visibility, first-time submission, reasoned outcomes, normal versus overdue waiting, and accessible operation. We use a request dashboard as the hub. Each card carries its stage, responsible office and timing. Details bring the reason and next action together. New Request and Help stay in the main navigation. The displayed date is fixed at 4 November 2026 so the demonstration remains repeatable.”

1 00 to 3 00 Complete core workflow

1:00-1:25: Open Guest Speaker Session, #E-031, W3. Point to Mentor, Faculty Mentor, last update and 6 November expected date. Expand history.

Narration: “This request is with the Faculty Mentor. Its review is expected by 6 November, so it is on time. We can identify the stage and responsible office without contacting anyone. The history shows the recorded handoff, while the last-update timestamp makes the age of that information visible.”

1:25-1:50: Return to My Requests; open Annual TechFest, #E-024, W4. Point to two-day overdue badge and simulated escalation disclosure.

Narration: “TechFest is different. It is with the Department Office and was expected by 2 November, two days before the demo clock. The word Overdue and elapsed days carry the distinction without colour. The escalation message is explicitly simulated: no supervisor notification is actually sent.”

1:50-2:15: Expand Instructor demo controls, choose Changes needed and Apply simulated update. Close controls; show W5 reason.

Narration: “This control simulates a staff update. The department asks for changes because Demo Hall A has an examination on 22 November. The reason tells Hira what she can do: choose a different date or venue. This staff decision is fictional; navigation, editing and state changes in the browser are implemented.”

2:15-2:45: Edit and resubmit W6. Choose 23 November 2026, Demo Hall B and a short reviewer note. Submit. Pause on loading and W8 confirmation.

Narration: “Hira edits the existing request rather than starting again. The reason stays above the form. She changes the date and venue and submits. During processing, the action is disabled and shows a loading label. The receipt keeps the request ID and states the new Department review stage and expected date, 7 November.”

2:45-3:00: Open updated details; expand history. Return to the hub and briefly show the separate Refused sample #E-018.

Narration: “The updated details and history preserve the correction. A separate refused request explains that an institutional sponsor was missing, so a refusal also has a stated reason.”

3 00 to 4 00 Error path and recovery

3:00-3:20: Apply Changes needed to #E-024 again. Edit, clear date and venue, type “Please retain my note”, and submit.

Narration: “For the alternate path, we deliberately leave date and venue empty. Submission is blocked. The summary says nothing was sent, focus moves to it, and the note we entered remains. Each affected field has its own specific message.”

3:20-3:40: Use an error-summary link to focus the date; correct to 24 November and Demo Hall B; submit and show confirmation.

Narration: “The summary links take us directly to the fields that need attention. We correct them and submit again. The form succeeds without making us repeat the information already supplied.”

3:40-4:00: Load empty dashboard; open New Request W9; show required guidance and Help link. Do not submit another long example.

Narration: “The empty dashboard leads to a working creation form. Required information is explained before submission. In this slice an event-plan field replaces supporting documents; uploads and real university document rules are outside the demonstration.”

4 00 to 4 40 Accessibility evidence and limits

Shot: Show keyboard focus on the skip link and a field; show text statuses in grayscale or the captured mobile grayscale evidence. Briefly show the evidence summary with exact current test counts. Use only claims still true for the submitted version.

Narration: “The build has native keyboard controls, a visible focus outline, a skip link, labelled fields and error associations. Route changes focus the main heading; dynamic notices use a polite status region. Our recorded automated run passed eighteen browser scenarios and twenty-four axe snapshots without reported A or AA violations. Those checks establish behavior within their scope. They do not establish spoken comprehension, usability success rates or complete accessibility conformance. The team walkthrough form captures the actual keyboard and screen-reader observations required by the brief.”

4 40 to 5 00 Focused feedback question

Shot: Restore samples and show W1 with #E-024 and #E-031 visible together. End by 4:55 if possible.

Narration: “Our focused feedback question is: Can users distinguish an approval request that is progressing normally from one that is overdue without needing further explanation? Please focus on the status words, expected dates and placement of the timing information.”

Submission readiness and checkpoint

Prepared artifacts are ready for team review; the following status separates them from completed activity evidence.

Brief item	Current artifact or next action
Proposed IA, navigation rationale and traceability	Prepared in Part A; feature-sort pack and results form supplied
Card-sort-derived final IA	Complete and attach the feature-sort record, then reconcile the navigation
Connected grayscale screens and error path	Working W1-W10 HTML and preserved original W1-W8
Hierarchy and Gestalt notes	Per-screen notes and wireframe review supplied for team review
Keyboard and screen-reader audit logs	POUR procedures and issue forms supplied; enter the team's actual observations
Fixed versus deferred issues	Engineering corrections recorded separately; add walkthrough findings and retest status
Studio critique record	Recording form and review checklist supplied; attach any session notes
Functional core task and creation	Implemented in index.html; automated browser evidence attached
Design system, scope and simulation notes	Prepared in Part B Implementation
Loading, disabled, empty, error and success	Implemented and recorded in automated evidence
Accessibility carry-forward	Native controls, focus and error semantics implemented; complete the team re-check
AI disclosure and reflection	Disclosure drafted with review fields; team to complete review record and both reflection answers
MP4 and captions	Script and shot list prepared; record original team narration and caption the final cut
Video name, duration and permissions	Export D08-Shmeezki&Mahid-D4-Prototype.mp4, at most five minutes; verify sharing and local playback
Live checkpoint	Run the submitted file and ensure all members can explain the code and simulation boundaries

Launch and freeze procedure

Open index.html, load samples and repeat the README demonstration. After code changes, repeat verification and regenerate matching evidence. Confirm that the video shows the submitted version. Retain the reviewed source, logs and locally playable video for the checkpoint and D5A inspection.